A Fibonacci Spine in a Five-fold Chorus of Ogdoecontahedron Packings



Five periodic packings of one icosahedral solid meet round a five-fold axis; the copies that must be removed along the axis form, exactly, the canonical Fibonacci cut-and-project chain, its window fixed by a five-fold ring clashing with itself, until a tie at height 12.8

DRAFT 1 · 15 October 2026 · not for release

DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com

DICTO DOI: on publication · Licence CC BY 4.0 · Data, scripts, exact checks and renders: https://github.com/DICTOR-Master/kaleidohedra (branch research, studies/2026-10-16-ocd-pyrochlore, steps 17–29).

The Ogdoecontahedron, its pyrochlore attachment networks, the five-fold "frog chorus" of five voices, and the questions that led to the exact chorus, its boundary, the shape of its core and the fork are DICTO's. Computation, exact arithmetic, renders and the literature search were done with Claude (Anthropic) as a research assistant; one bounded literature review was done with Claude Fable.

Abstract. The DICTO Ogdoecontahedron is a convex icosahedral solid with 20 regular hexagons and 60 kites. Its hexagon directions contain five inscribed cubes, and each cube gives a periodic packing in which copies meet face to face on hexagons. We place the five packings ("voices") in five 72° wedges round a five-fold axis, which is the icosahedral irrational twin, and remove clashing copies in whole five-fold rings. Checked exactly in $\mathbb{Q}(\sqrt{5})$, the result is overlap-free to radius 10 (2756 sites, 385 removed, 7465 close pairs). Every copy lies at a golden height $(m + n\varphi)u$ along the axis. The copies removed near the axis lie in one plane on a rank-2 lattice with axis slope $-\varphi$, and they are exactly the canonical singular Fibonacci cut-and-project chain, window $[0, L)$ with $L^2 = 64/81 + 128\sqrt{5}/405$, the central copy being one of its points. The window has a mechanism: a ring of five copies overlaps itself exactly when it is closer to the axis than t_c, $t_c^2 = 8/9 + 8\sqrt{5}/45$. Since $t_c < L$ and no lattice point lies between them below height 12.81, the chain is exactly Fibonacci there. At height 12.81 the first such point clashes with exactly one other copy, and the two equally minimal removals give either the Fibonacci chain or a thinned three-gap chain: hardness alone stops deciding. Irrational interfaces as one-dimensional quasicrystals, model sets near irrational lines and hard-core windows are known; we did not find this selection along a five-fold junction line, its self-clash window or the fork, and present them as a new exact instance of known ideas.

1. Introduction

Periodic crystals repeat; quasicrystals are ordered without repeating, and can be described as cuts through higher-dimensional lattices. Between the two lie twinned crystals. When grains of one crystal meet at an irrational orientation, the interface between them can itself be a one-dimensional quasicrystal (Sutton 1988, 1989, 1992; Rivier & Lawrence 1988; Gratias & Thalal 1988), and five cubic variants turned by 72° about an irrational axis form the icosahedral irrational twin, whose coincidence-lattice approximants stack in Fibonacci order (Ranganathan, Srivastava & Lord 2006). Pauling (1985) proposed, wrongly, that icosahedral

quasicrystals are multiply twinned cubic crystals; Prodan et al. (2017) describe icosahedral tilings as cyclic twinning of golden rhombohedra.

This paper follows one solid through such a twin, exactly. Five periodic packings of the DICTO Ogdoecontahedron meet round a five-fold axis. Hard copies cannot overlap, so some must be removed ("silenced"). We show that the copies removed along the axis are not merely quasiperiodic but are exactly the canonical Fibonacci chain, that the window comes from a five-fold ring of copies clashing with itself, and that beyond a height of 12.81 the minimal removal is no longer unique.

Every statement is labelled. **Exact** means proved or checked in $\mathbb{Q}(\sqrt{5})$ with rational arithmetic; **float** means computed in floating point and not proved.

2. The solid

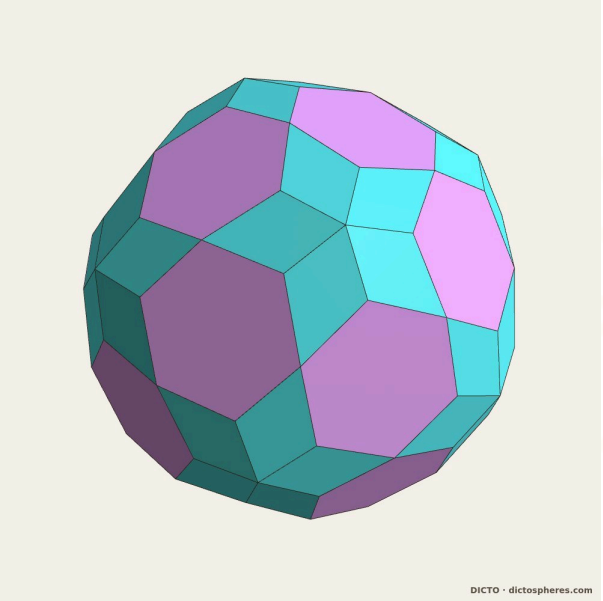


Fig. 1. The DICTO Ogdoecontahedron: 20 regular hexagons and 60 kites.

The DICTO Ogdoecontahedron (DICTO's companion paper, in draft) is the polar reciprocal of the DICTO Truncated-Icosahedral Hecatodihedron (DICTO DOI: 10.5281/zenodo.23360858), and the second station of the Spheres of Dharma turned inside out (DICTO DOI: 10.5281/zenodo.23310833). It has 80 faces (20 regular hexagons, 60 kites with angles 117.85°, 68.62°, 55.69°), 102 corners (12 of degree 5, 30 of degree 4, 60 of degree 3) and 180 edges. All coordinates are exact in $\mathbb{Q}(\sqrt{5})$. It is not canonical: its edges lie at two distances from the centre, so it has no midsphere.

Face arrangement (exact). An exact face-map comparison (mirror images allowed, with positive and negative controls) shows that its face arrangement is that of the pentakis dodecahedron with its 20 six-fold corners cut down to the edge midpoints, equivalently the dual of L0(D), the joined-lace dodecahedron in Antiprism's operator notation. The arrangement is therefore obtainable by standard operations. It is not, however, in David McCooey's *Visual Polyhedra*, whose 553 solids we scanned in full; its nearest relative there is the 6-truncated pentakis dodecahedron (20 hexagons, 60 pentagons), whose truncation ends, at the edge midpoints, in this arrangement. The printed catalogues (Wenninger, Holden, Pugh) were ruled out by scope, not read. The metric solid, with regular hexagons and these kites, was not found in any source checked.

3. Five voices and the chorus

Bonds and cubes (exact). Joining two copies face to face on a hexagon, with no rotation, displaces one by the bond vector $2d \cdot n/|n|^2$ of that hexagon (n its normal, d its plane offset). The 20 hexagon directions contain exactly five sets of eight that form a cube (pairwise cosines $\pm 1/3$ and -1): the five inscribed cubes. Each cube's bond vectors generate a rank-3 lattice, and copies on it meet face to face on hexagons, with joints on a pyrochlore lattice. These five periodic packings are the five **voices**.

The chorus. Fix a five-fold axis through a degree-5 corner of the central copy. Space is cut into five 72° wedges round the axis (starting angle 59°, a choice), and wedge k is filled with voice k's lattice points; points on the axis are shared. Turning by 72° about the axis carries each voice to the next, so the arrangement is five-fold symmetric. Neighbouring voices are related by 44.4775° about a shared three-fold axis: this is the icosahedral irrational twin (Ranganathan et al. 2006). Where wedges meet, and near the axis, copies of different voices overlap. They are removed in whole five-fold rings by a greedy rule (float clash test): repeatedly remove the ring of the copy with most clashes, ties going to the copy farther from the centre.

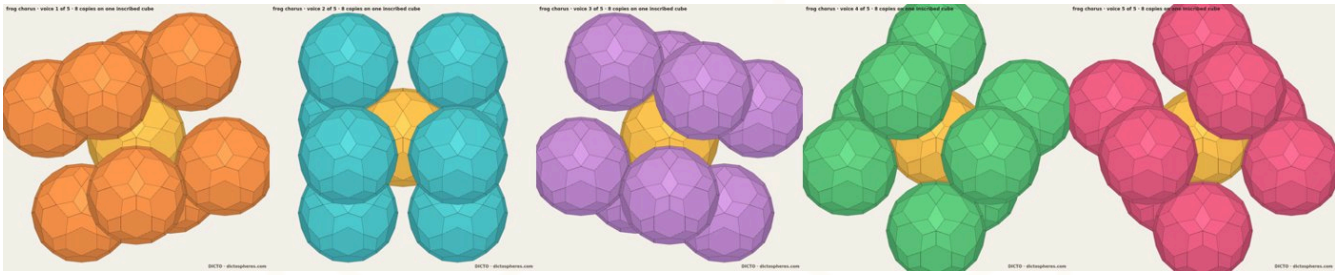
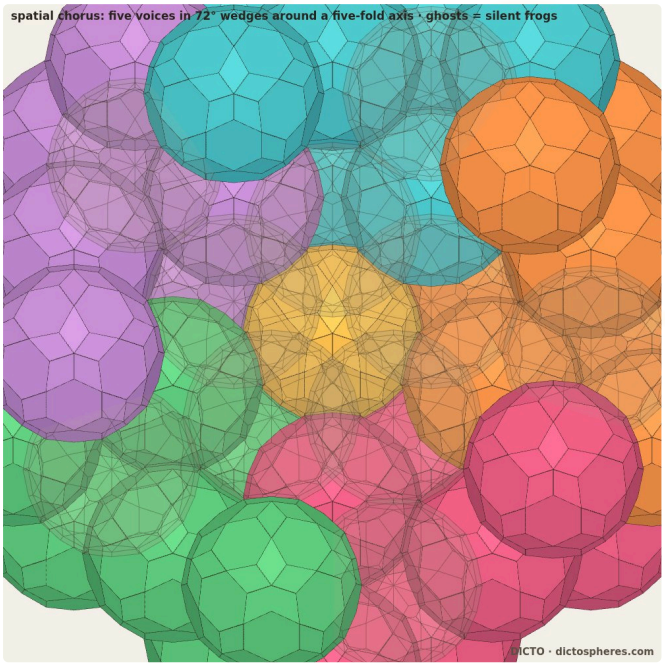


Fig. 2. The five voices: on each inscribed cube, eight copies joined face to face on hexagons round a central copy.



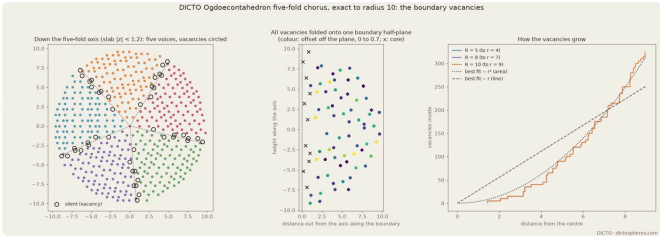


Fig. 3–4. (3) The chorus down the five-fold axis, voices in five colours, removed copies as ghosts (small early build). (4) The radius-10 build: vacancies circled; vacancies folded onto one boundary half-plane; growth of the vacancy count with radius.

4. Exactness

Two copies at X and Y are apart if and only if some face plane $n \cdot x = d$ of the solid has $n \cdot (X - Y) \geq 2d$ (the difference body of a centrally symmetric solid K is 2K); they touch when the best face gives equality. Copies farther apart than twice the circumradius cannot overlap, so checking every closer pair is complete. All positions are integer combinations of exact bond vectors.

Radius	Sites	Removed	Close pairs checked	Overlaps	Touching
5	336	80	620	0	620
8	1471	240	3630	0	3605
10	2756	385	7465	0	7390

The choice of what to remove is float; the result is exact. At radius 5 an integer program (float clash test) finds the minimal five-fold-symmetric removal to be 16 rings (80 copies), the same as the greedy rule, and not unique; without symmetry the minimum is 72.

Lemma (golden heights, exact). Measured along the five-fold axis, every bond vector has height $(m + n\varphi)u$ with $u^2 = 8/81 - 8\sqrt{5}/405$ ($u \approx 0.233658$), the pairs (m, n) being only $(\pm 1, 0)$ and $\pm(1, 2)$. Every site is an integer combination of bonds, so **every copy in the chorus, at any radius, sits at a golden height $(m + n\varphi)u$.**

5. The core is the canonical Fibonacci chain

Near the axis the removals come in rings of five, one copy per wedge. In the radius-10 build there are 11 such rings. Their squared distances from the axis include φ^2, φ^4, φ^6 and φ^8 times the innermost (exact).

One plane, one lattice, one window (exact). Ten of the 11 rings have their wedge-0 copy in one plane through the axis. The 73 sites of that plane in the build are integer combinations of two basis vectors, and in that basis the axis has slope $-\varphi = -1/2 - \sqrt{5}/2$, which is irrational, so heights along the axis are all distinct. With t the distance across from the axis, the removed plane copies are exactly those with t below a window edge: there is no kept copy inside and no removed copy outside. Lattice points of a planar lattice in a strip along an irrational line are, by definition, a cut-and-project (model) set (Baake & Grimm 2013).

The canonical window (exact). The gaps between consecutive removed heights take the values S = 1.22345, L = 1.97958 and T = 3.20304, with $L/S = \varphi$ and $T = L + S$ exactly; the gap vectors of L and S have plane coordinates $(1, -1)$ and $(1, 0)$, a lattice basis. The canonical window length is therefore $L_{can}^2 = 64/81 + 128\sqrt{5}/405$ ($L_{can} = 1.223451$), and the single T is where the chain point at height 0, the **central copy**, sits (on the axis, t = 0, kept). With window $[0, L_{can})$, the chain points inside the build are exactly the 11 removed copies and the centre; the copy exactly on the far edge ($t = L_{can}$, height -0.756) is kept. Extended exactly to heights ± 45, the same window gives 54 points with only two gaps in ratio φ (33 long, 20 short) forming a factor of the Fibonacci word. Both window edges are lattice points: the chain is a **singular** member of the Fibonacci family, with the centre on the closed edge.

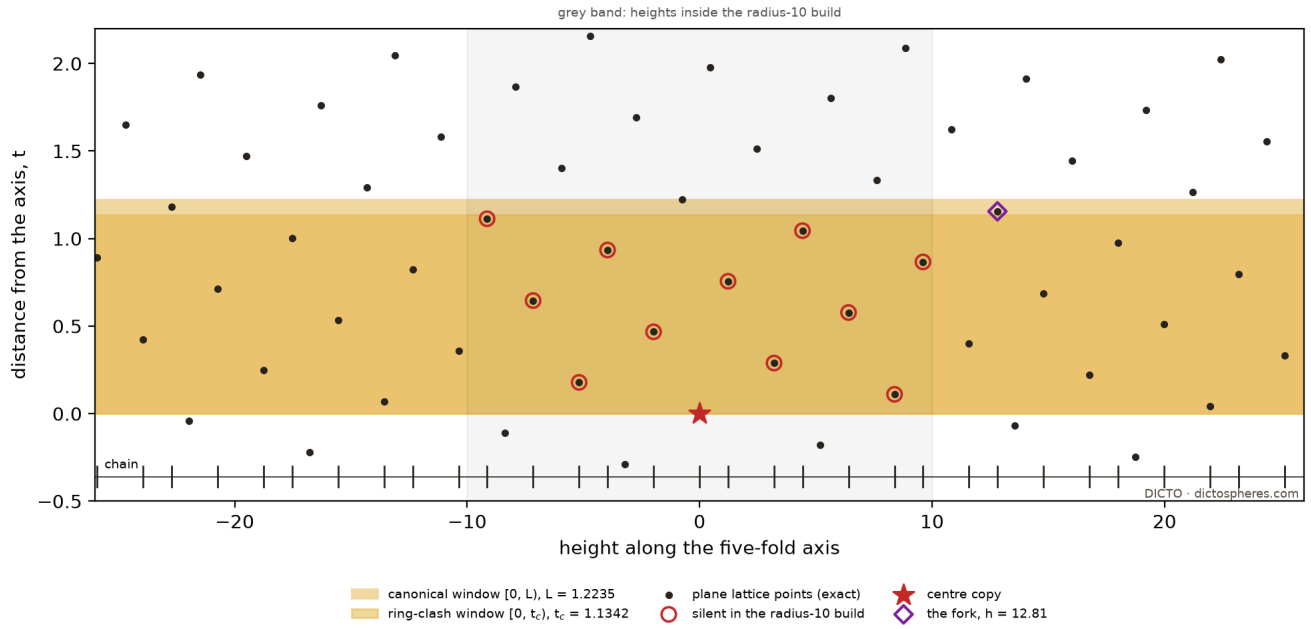


Fig. 5. The core plane as a cut-and-project picture (exact points, drawn in float). Dots: plane lattice points by height along the axis and distance t from it. Circled: removed in the radius-10 build (grey band). Pale band: canonical window [0, L); darker band: ring-clash window [0, t_c). Star: the central copy. Diamond: the fork at height 12.81. Ticks below: the canonical chain.

6. Why that window: the ring clashes with itself

A copy at distance t from the axis and its four ring-mates (the same copy turned by 72°k, sites of the other voices) differ by vectors perpendicular to the axis and proportional to t. So whether a ring overlaps itself depends on t alone. Checked exactly for every plane copy with t < 1.6: each removed copy overlaps its ring-mates, each kept one clears them. The critical distance, where ring-mates just touch, is exact and the same for every plane site:

$$\mathbf{t_c^2 = 8/9 + 8\sqrt{5}/45, \quad t_c = 1.134201.}$$

It lies strictly between the largest removed t (1.113132) and L_{can} (1.223451); the copy on the canonical edge clears its ring-mates by $28/81 - 4\sqrt{5}/27 \approx 0.014$ (best face separation). So $t_c \neq L_{can}$, and hardness alone gives the narrower window [0, t_c). No plane lattice point has $t_c \le t < L_{can}$ below height 12.812, so up to there the two windows select the same points and **the core is exactly the canonical Fibonacci chain, beyond the build's own reach (|h| < 10)**. The first lattice points between the two windows are at heights 12.812, −22.71, 34.77 and −44.66.

7. The fork at height 12.81

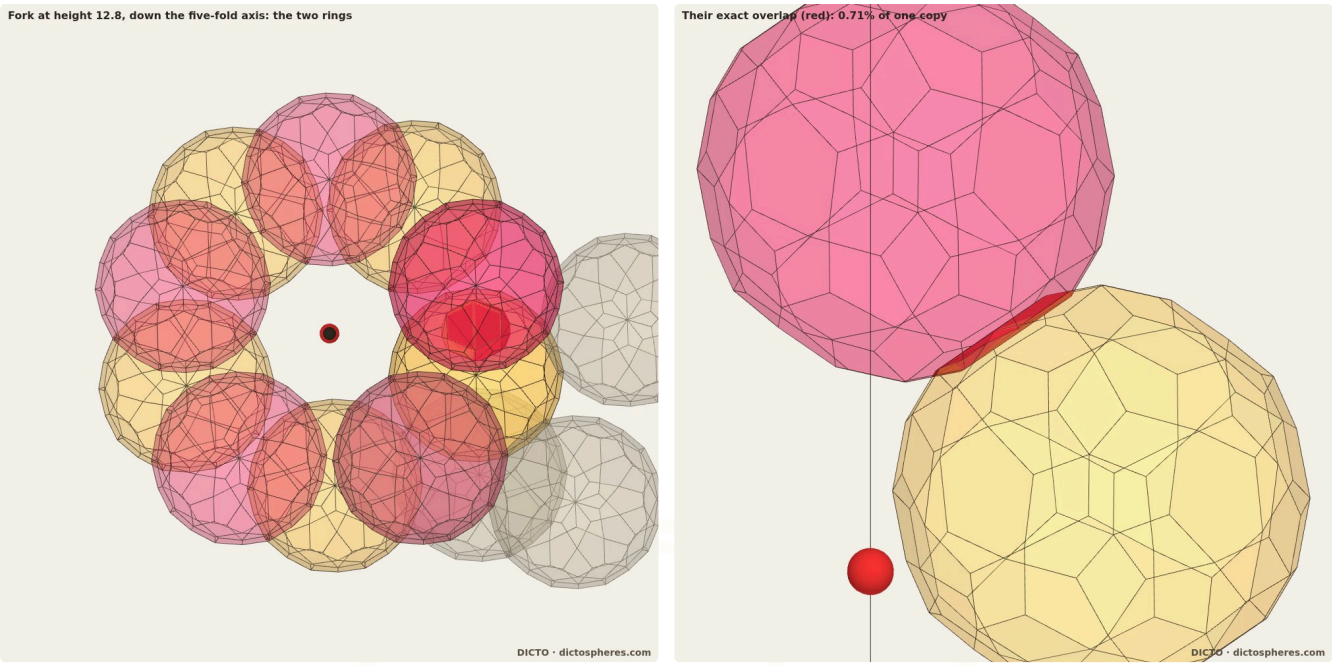


Fig. 6–7. (6) Down the axis: the ring through the core copy at height 12.81 (gold) and the ring at 13.80 (magenta); red, where a gold and a magenta copy overlap. (7) Close-up: the exact overlap, a thin sliver of 0.71% of one copy (float volume). A supplementary movie shows both choices.

We built every site of all five voices in a cylinder round the axis at that height (1135 sites; exact). The core copy at height 12.8121 (t = 1.1553, plane coordinates (8, −4)) clears all four ring-mates, as §6 predicts. Of the seven copies near enough to matter, it overlaps **exactly one**: a copy at height 13.8019, distance 1.1690 from the axis, in the next wedge, outside the core plane. That copy's own ring does not clash with itself, so nothing else forces it out. The deepest face separation between the two is $232/81 - 104\sqrt{5}/81 \approx -0.0068$ (exact, negative: overlap). The target also touches four other copies exactly.

So one of the two rings must go, and either single removal resolves the clash:

- **remove the core ring**: height 12.81 joins the chain, and the spine stays the canonical Fibonacci chain;
- **remove the ring at 13.80**: the core copy stays, height 12.81 leaves the chain, and the spine becomes a thinned chain with gaps L, S and L + S.

Both rings have the same number of clashes. The chorus's greedy rule breaks the tie by distance from the centre and removes the ring at 13.80, taking the thinned chain (float, on the 1135-site patch). **Beyond height 12.81 hardness alone no longer decides the spine**: pure Fibonacci continues only under a rule that prefers removing copies in the core plane. Either way the spine stays a one-dimensional cut-and-project chain. We describe this as a degeneracy of the minimal removal, not as a new phenomenon; the closest prior idea we found is Sutton's (1989) energetically degenerate variants of an irrational interface, separated by line defects.

8. The seams (float)

Away from the axis, removals lie along the five wedge boundaries. Their number grows like area, not length (5, 20, 60, 95, 145, 205 and 275 within radius 3 to 9), and their density per boundary area is still rising at radius 9 (0.31 to 0.43 from radius 5 to 9). The distance from

the seam plane does not decide which copies go: kept and removed copies share offsets on both sides, so there is no simple slab rule. The 66 distinct vacancies of one seam diffract strongly: strongest peak I/N = 46.6, against 66 for a perfect Bragg peak and 11.8 ± 2.0 for random sets of the same size, with the strongest peaks within the peak width of projected reciprocal-lattice vectors of the two neighbouring voices. This is consistent with incommensurate, quasiperiodic-type order, but 66 points cannot tell it from a long-period approximant.

9. Discussion

Relation to known work. That interfaces between periodic grains can be one-dimensional quasicrystals is classical (Sutton; Gratias & Thalal; Rivier & Lawrence), as is the fact that lattice points near an irrational line form a model set (Baake & Grimm 2013). Katz & Gratias (1993) bounded icosahedral windows by hard-core and closeness conditions: there the window is *chosen* to maximise density inside a six-dimensional model, and exclusion acts between points one lattice shift apart. Here nothing is chosen: the removed set is the window, forced by overlap, and the exclusion acts between a copy and its own five-fold rotated images, which belong to four other periodic packings. One-dimensional lattice gases have Sturmian (cut-and-project) ground states (Hubbard 1978; van Enter, Koivusalo & Miękisz 2019), a related selection by energy rather than by shape.

Not a disclination. Five-fold twinned particles (fivelings) carry a wedge disclination along their axis, with an angle gap relieved by strain or defects (Hofmeister 1998). The chorus has no gap and no strain; its axis is a line of irrational direction shared by five lattices, and the price of five-fold symmetry is paid in vacancies. Hard shapes are known to form five-fold twinned clusters in simulation (Lee & Glotzer 2022), in floating point and with an angle gap; we know of no exact junction check for a hard polyhedron.

What we did not find. A Fibonacci chain *along* a five-fold junction line rather than across a planar interface; a window fixed by a ring clashing with itself, with an exact critical distance; an exact overlap check of a five-fold junction of hard-polyhedron packings; a tie between two windows. The search covered our reference library, the web and one bounded literature review (§11). Several key papers are closed access and were read through abstracts and open restatements only; we present the results as a new exact instance of known ideas and make no claim of priority.

Limits. The packing is checked to radius 10 and the fork on a 1135-site patch; the choice of removals is float; the seams are float throughout; the off-plane ring at distance 1.0042 from the axis belongs to a second family not analysed here.

10. Open questions

- Is there a natural local rule under which the spine stays pure Fibonacci at every height, and what does the chorus do at the later departure points (−22.71, 34.77, −44.66)?
- Are the seams model sets, and is their density per area finite in the limit?
- What is the off-plane family of core rings?
- Which other solids have five inscribed cubes in their bond directions, and do their choruses give other chains?

11. Reproducibility

Scripts and outputs are in studies/2026-10-16-ocd-pyrochlore of the Kaleidohedra repository (branch research): `chorus_exact_big.py` (the exact chorus, any radius), `chorus_heights_exact.py` (§4), `chorus_modelset.py` and `chorus_canonical.py` (§5), `chorus_window_why.py` (§6), `chorus_h128.py` and `chorus_fork_data.py` (§7), `chorus_boundary.py` (§8), `ocd_dl0d.py` and `ocd_catalogue_scan.py` (§2), each with its output in data/. The literature review is `FABLE-PRIORART.md`. Movies of the boundary sweep and of the fork are supplementary.

References

1. M. Baake, U. Grimm, *Aperiodic Order, Vol. 1: A Mathematical Invitation*, Cambridge University Press (2013). doi:10.1017/CBO9781139025256
2. D. Gratias, A. Thalal, Hidden symmetries in general grain boundaries, *Phil. Mag. Lett.* 57 (1988) 63–68. doi:10.1080/09500838808229611
3. H. Hofmeister, Forty years study of fivefold twinned structures in small particles and thin films, *Cryst. Res. Technol.* 33 (1998) 3–25.
4. J. Hubbard, Generalized Wigner lattices in one dimension and some applications to TCNQ salts, *Phys. Rev. B* 17 (1978) 494–505. doi:10.1103/PhysRevB.17.494
5. A. Katz, D. Gratias, A geometric approach to chemical ordering in icosahedral structures, *J. Non-Cryst. Solids* 153–154 (1993) 187–195. doi:10.1016/0022-3093(93)90340-4
6. S. Lee, S. C. Glotzer, Entropically engineered formation of fivefold and icosahedral twinned clusters of colloidal shapes, *Nat. Commun.* 13 (2022). doi:10.1038/s41467-022-34891-5
7. D. McCooey, *Visual Polyhedra*, dmccooey.com/polyhedra (accessed 15 October 2026).
8. L. Pauling, Apparent icosahedral symmetry is due to directed multiple twinning of cubic crystals, *Nature* 317 (1985) 512–514. doi:10.1038/317512a0
9. A. Prodan, R. Dušić Hren, M. A. van Midden, H. J. P. van Midden, E. Zupanič, The equivalence between unit-cell twinning and tiling in icosahedral quasicrystals, *Sci. Rep.* 7 (2017) 12474. doi:10.1038/s41598-017-12669-w
10. S. Ranganathan, A. K. Srivastava, E. A. Lord, Coincidence-site lattices as rational approximants to irrational twins, *J. Mater. Sci.* 41 (2006) 7696–7703. doi:10.1007/s10853-006-0966-y
11. N. Rivier, A. J. A. Lawrence, Quasicrystals at grain boundaries, *Physica B+C* 150 (1988) 190–202. doi:10.1016/0378-4363(88)90122-2
12. A. P. Sutton, Irrational tilt grain boundaries as one-dimensional quasicrystals, *Acta Metall.* 36 (1988) 1291–1299. doi:10.1016/0001-6160(88)90281-7
13. A. P. Sutton, Quasiperiodicity in irrational interfaces, *Phase Transitions* 16 (1989) 563–574. doi:10.1080/01411598908245732
14. A. P. Sutton, Irrational interfaces, *Prog. Mater. Sci.* 36 (1992) 167–202. doi:10.1016/0079-6425(92)90008-U
15. A. C. D. van Enter, H. Koivusalo, J. Miękisz, Sturmian ground states in classical lattice-gas models, *J. Stat. Phys.* 178 (2019) 832–844. doi:10.1007/s10955-019-02464-4
16. S. Xiong, W. Zou, P. Zhang, K. Jiang, Framework for quasiperiodic interfaces: proximal coincidence point set and computation, arXiv:2603.21158 (2026).
17. DICTO, The DICTO Truncated-Icosahedral Hecatodihedron. DICTO DOI: 10.5281/zenodo.23360858
18. DICTO, Spheres of Dharma. DICTO DOI: 10.5281/zenodo.23310833
19. DICTO, The DICTO Ogdoecontahedron (draft, 16 October 2026).